Disconnected qTMD/PDF One-Point Building Blocks

Stochastic Quark Loops for Pion and Proton qTMD Analysis

PyQUDA Measurement Notes

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Contents

1 Purpose

This note documents the first PyQUDA implementation of disconnected qTMD/PDF one-point quark loops. The source file is `pyquda_measurement_utils/Disconnected_1pt_qTMD_vibe_develop.py`, and the Perlmutter application is `application/qTMD_disconnected_1pt/perlmutter`.

The loops are hadron independent. Pion and proton information enters later, when the loop is combined with the corresponding hadron two-point function in ensemble analysis.

2 Disconnected qTMD Loop

For a stochastic source ξ_r, define

$$\eta_r = D^{-1}\xi_r, \tag{1}$$

where D is the same clover Dirac operator used by the connected qTMD measurements. The implemented disconnected qTMD/PDF loop is

$$L_{\Gamma,b}(q,\tau) = \frac{1}{N_r}\sum_{r=1}^{N_r}\sum_{\mathbf{x}} e^{i\mathbf{q}\cdot\mathbf{x}}\ \mathrm{Tr}_{sc}\left[\eta_r^\dagger(x)\,\Gamma\,O_b\,\xi_r(x)\right], \tag{2}$$

where $x = (\mathbf{x}, \tau)$, Γ is one of the 16 bilinear gamma matrices, and O_b is a nonlocal displacement or Wilson-line operator.

The current code convention is therefore

$$\boxed{\text{loop convention} = \eta^\dagger \Gamma O_b \xi}. \tag{3}$$

The nonlocal operator O_b acts on the stochastic source side. This choice is intentionally minimal and makes the first implementation easy to compare against connected PDF and local-current limits.

3 Supported Operators

The first implementation supports three operator kinds:

`operator_kind`	Operator	Status
`GI_PDF`	Straight-z gauge-invariant PDF Wilson line	Implemented
`CG_PDF`	Straight-z coordinate-gauge displacement	Implemented
`CG_qTMD`	qTMD-style coordinate-gauge displacement	Implemented

3.1 Straight-z PDF Operators

For PDF operators,

$$b = b_z\hat{z}. \tag{4}$$

The coordinate-gauge version is a plain lattice shift:

$$O_b^{\mathrm{CG\_PDF}}\xi(x) = \xi(x + b_z\hat{z}). \tag{5}$$

The gauge-invariant version applies the straight Wilson line through repeated covariant shifts:

$$O_b^{\mathrm{GI\_PDF}}\xi(x) = U_z(x)U_z(x+\hat{z})\cdots\xi(x + b_z\hat{z}), \tag{6}$$

with the corresponding backward-link product for negative b_z. In code this is implemented by repeated calls to `gauge.pure_gauge.covDev`.

3.2 Coordinate-Gauge qTMD Operator

The current qTMD operator follows the connected qTMD diagnostic convention:

$$O_b^{\mathrm{CG\_qTMD}}\xi(x) = \xi(x + b_T\hat{e}_\perp + b_z\hat{z}), \tag{7}$$

where $\hat{e}_\perp$ is either $\hat{x}$ or $\hat{y}$. The two transverse directions are stored as `b_X` and `b_Y` in the HDF5 output.

3.3 Gauge-Invariant qTMD Staple Operator

The production code also supports a fixed-length gauge-invariant staple operator, `operator_kind=GI_qTMD`. Its Wilson index is $[b_T, b_z, \eta, \perp]$, with even physical b_z, non-negative η, and $\eta \geq |b_z|/2$.

4 Momentum Convention

The loop momentum is the current momentum transfer,

$$q = p_f - p_i. \tag{8}$$

The one-point loop itself has no hadron source or sink momentum. Those labels enter when the loop is combined with a pion or proton two-point function.

The code uses `pyquda_utils.phase.MomentumPhase`, whose convention is

$$\Phi_q(x; x_0) = \exp\left[+2\pi i \sum_{j=x,y,z} \frac{q_j(x_j - x_{0,j})}{L_j}\right]. \tag{9}$$

The disconnected loop is projected as

$$L_{\Gamma,b}(q,\tau) = \sum_{\mathbf{x}} \Phi_q(\mathbf{x};\mathbf{x}_0)\, \ell_{\Gamma,b}(\mathbf{x},\tau), \tag{10}$$

so a local test field with plane-wave dependence

$$\ell_{\Gamma,b}(\mathbf{x},\tau) \propto \exp\left[-2\pi i \sum_{j=x,y,z} \frac{q_j(x_j - x_{0,j})}{L_j}\right] \tag{11}$$

is selected by the $+q$ projection. This is the same Fourier sign used in the connected qTMD/EMT application layers; the physical transfer label remains $q = p_f - p_i$.

5 Combination with Hadron Two-Point Functions

For a hadron two-point correlator $C_2^H(p_f, t)$, the disconnected qTMD three-point building block is formed as

$$C_{3,\Gamma,b}^{H,\text{disc}}(p_f, p_i; t, \tau) = \left\langle C_2^H(p_f,t)\, L_{\Gamma,b}(q,\tau)\right\rangle_{\text{cfg}} - \left\langle C_2^H(p_f,t)\right\rangle_{\text{cfg}} \left\langle L_{\Gamma,b}(q,\tau)\right\rangle_{\text{cfg}}, \tag{12}$$

with

$$q = p_f - p_i. \tag{13}$$

The corresponding ratio is typically

$$R_{\Gamma,b}^{H,\text{disc}}(t,\tau) = \frac{C_{3,\Gamma,b}^{H,\text{disc}}(p_f,p_i;t,\tau)}{C_2^H(p_f,t)}, \tag{14}$$

before applying the qTMD/PDF renormalization and soft-factor analysis.

The vacuum-subtraction term is essential because the one-point loop is hadron independent.

6 Stochastic Sources and Hierarchical Probing

The source scheme is controlled by

$$\texttt{noise\_scheme} \in \{\texttt{zn}, \texttt{hierarchical\_probing}\}. \tag{15}$$

For hierarchical probing, a base source ξ_b is multiplied by Rademacher probing vectors $h_k(x) = \pm 1$:

$$\xi_{b,k}(x) = h_k(x)\xi_b(x). \tag{16}$$

The number of effective inversions is

$$N_{\text{eff}} = N_{\text{base}}\, N_{\text{HP}}. \tag{17}$$

The implemented HP ordering choices are

Ordering	Meaning
`global_xyzt_gray_projected_to_evenodd`	Gray code the full x, y, z, t site id.
`spatial_xyz_then_t_gray_projected_to_evenodd`	Gray code spatial sites first, then append time bits.

Spin-color dilution and time dilution are not currently implemented.

7 HDF5 Layout

The output tag helper is `get_disconnected_qTMD_1pt_file_tag`. The writer stores:

$$\texttt{raw/loop\_pervec} \quad \text{with shape} \quad [N_{\text{eff}}, N_W, N_\Gamma, N_q, N_t]. \tag{18}$$

The raw bookkeeping datasets are

$$\texttt{raw/source\_index}, \quad \texttt{raw/base\_noise\_index}, \quad \texttt{raw/hp\_index}. \tag{19}$$

The averaged data are divided by the spatial volume and stored in a qTMD-like layout:

$$\texttt{avg/SS/<gamma>/PX...PY...PZ.../b\_X/eta0/bT.../bz....} \tag{20}$$

For y-direction transverse displacements the group is `b_Y`.

8 Validation Status

The first implementation has been checked with:

- Python syntax compilation.
- A Perlmutter login-node smoke test for `GI_PDF` at $b_z = 0$.
- A Perlmutter login-node smoke test for `GI_PDF` at $b_z = 1$.
- A Perlmutter login-node smoke test for `CG_qTMD` at $b_T = 1, b_z = 1$.

8.1 Local/PDF Limit Sanity Checks

At zero displacement and zero momentum transfer, the different disconnected operator implementations should collapse to the same local bilinear loop:

$$O^{\text{GI\_PDF}}_{b_z=0} = O^{\text{CG\_PDF}}_{b_z=0} = O^{\text{CG\_qTMD}}_{b_T=0,b_z=0} = 1. \tag{21}$$

Therefore, for the same gauge field, stochastic source, gamma matrix, and momentum $q = 0$, the expected equality is

$$L^{\text{GI\_PDF}}_{\Gamma,b_z=0}(0,\tau) = L^{\text{CG\_PDF}}_{\Gamma,b_z=0}(0,\tau) = L^{\text{CG\_qTMD}}_{\Gamma,b_T=0,b_z=0,\hat{x}}(0,\tau) = L^{\text{CG\_qTMD}}_{\Gamma,b_T=0,b_z=0,\hat{y}}(0,\tau). \tag{22}$$

This check is useful because it tests the HDF5 bookkeeping, Wilson-index ordering, gamma loop ordering, and the coordinate-gauge/GI local limits without requiring a hadron two-point function.

The current S8T32 Perlmutter login-node check used one stochastic vector with matched random seed and found exact agreement:

$$\max\left|L^{\text{GI\_PDF}} - L^{\text{CG\_PDF}}\right| = \max\left|L^{\text{GI\_PDF}} - L^{\text{CG\_qTMD}}_{\hat{x}}\right| = \max\left|L^{\text{GI\_PDF}} - L^{\text{CG\_qTMD}}_{\hat{y}}\right| = 0. \tag{23}$$

The same equality was verified for both `raw/loop_pervec` and the averaged $\Gamma = \gamma_5$ HDF5 path `avg/SS/5/PX0PY0PZ0/b_X/eta0/bT0/bz0`, with the y-direction branch stored under `b_Y` for `CG_qTMD`.

The regression helper is `tests/test_qtmd_disconnected_local_pdf_limit.py`. It skips automatically if the local HDF5 smoke-test files are not present, and otherwise requires exact equality for the local/PDF limit above.

8.2 Nonzero-b_z Shift Sanity Check

A second check probes the straight-z displacement convention at nonzero separation. For the coordinate-gauge operators, the PDF path at $b_T = 0$ should be exactly the $b_T = 0$ limit of the qTMD coordinate shift:

$$O^{\text{CG\_PDF}}_{b_z=\pm1} = O^{\text{CG\_qTMD}}_{b_T=0,b_z=\pm1,\hat{x}} = O^{\text{CG\_qTMD}}_{b_T=0,b_z=\pm1,\hat{y}}. \tag{24}$$

This equality checks the sign of the z-shift, the Wilson-index ordering, and the transverse-branch bookkeeping. The corresponding S8T32 check found exact agreement for both $b_z = +1$ and $b_z = -1$:

$$\max \left| L^{\text{CG\_PDF}}_{\Gamma,b_z=\pm1} - L^{\text{CG\_qTMD}}_{\Gamma,b_T=0,b_z=\pm1,\hat{x}/\hat{y}} \right| = 0. \tag{25}$$

The gauge-invariant PDF loop uses the same Wilson-index labels, $[0, 0, 0, 0]$, $[0, +1, 0, 0]$, and $[0, -1, 0, 0]$, but it is not expected to equal the coordinate-gauge loop on a nontrivial gauge field because the straight Wilson line is active:

$$O^{\text{GI\_PDF}}_{b_z=\pm1} = U_z \cdots \xi(x \pm \hat{z}) \neq \xi(x \pm \hat{z}) = O^{\text{CG\_PDF}}_{b_z=\pm1}. \tag{26}$$

The nonzero difference between `GI_PDF` and `CG_PDF` therefore confirms that the GI path is not accidentally falling back to the plain coordinate shift.

The regression helper is `tests/test_qtmd_disconnected_nonzero_bz.py`. During this check it also exposed and fixed a transverse-branch bug in the `CG_qTMD` operator: when switching from the $\hat{x}$ branch to the $\hat{y}$ branch, the shifted stochastic source must be reset to the local source before applying the new branch displacement.

8.3 Nonzero-q Phase Sanity Check

The regression helper `tests/test_disconnected_momentum_phase.py` uses a pure NumPy toy field to check the nonzero-momentum Fourier sign. For a tiny lattice and $q = (1, -2, 3)$, it verifies

$$\sum_{\mathbf{x}} \Phi_q(\mathbf{x}; 0)\, \Phi_q^*(\mathbf{x}; 0) = L_x L_y L_z, \tag{27}$$

while the opposite projection vanishes:

$$\sum_{\mathbf{x}} \Phi_{-q}(\mathbf{x}; 0)\, \Phi_q^*(\mathbf{x}; 0) = 0. \tag{28}$$

The same test also checks that changing x_0 only changes the expected overall phase, not the selected momentum magnitude.

9 Gauge-Invariant Staple qTMD Design

The `CG_qTMD` operator is only a coordinate-gauge diagnostic. The production `GI_qTMD` operator replaces the plain shift by a staple Wilson line. The convention is the fixed-staple-length convention used by the legacy GPT proton qTMD workflow. In the current notation, b_z is the physical final longitudinal separation and must be an even integer:

$$O^{\mathrm{GI\_qTMD}}_{b,\eta,\perp}\xi(x) = U_{\mathcal{C}(x;b,\eta,\perp)}\xi(x + b_T\hat{e}_\perp + b_z\hat{z}), \tag{29}$$

with

$$b_z \in 2\mathbb{Z}, \qquad \eta \geq \frac{|b_z|}{2}. \tag{30}$$

The three-segment staple path is

$$\mathcal{C}(x;b,\eta,\perp) : x \to x + \left(\eta + \frac{b_z}{2}\right)\hat{z} \to x + \left(\eta + \frac{b_z}{2}\right)\hat{z} + b_T\hat{e}_\perp \to x + b_z\hat{z} + b_T\hat{e}_\perp. \tag{31}$$

The signed segment lengths are

$$\left[\left(z, \eta + \frac{b_z}{2}\right), (\perp, b_T), \left(z, \frac{b_z}{2} - \eta\right)\right]. \tag{32}$$

For $\eta \geq |b_z|/2$, the geometric Wilson-line length is independent of b_z:

$$\left|\eta + \frac{b_z}{2}\right| + b_T + \left|\eta - \frac{b_z}{2}\right| = 2\eta + b_T. \tag{33}$$

This fixed length is useful for renormalization because Wilson-line self-energy effects are matched across different b_z at fixed η and b_T. With $b_T = 0$ and $\eta = |b_z|/2$, the staple reduces to the straight `GI_PDF` Wilson line. At $b_T = b_z = \eta = 0$, it reduces to the local operator.

The legacy GPT implementation used an equivalent convention with b_z^{legacy} denoting half of the final longitudinal separation:

$$b_z^{\mathrm{current}} = 2b_z^{\mathrm{legacy}}. \tag{34}$$

Its path comments and implementation correspond to $(\eta + b_z^{\mathrm{legacy}}) + b_T - (\eta - b_z^{\mathrm{legacy}})$, which is the same fixed-length staple after this relabeling.

The `GI_qTMD` implementation keeps the existing `CG_qTMD`, `CG_PDF`, and `GI_PDF` outputs unchanged. The first code-side helper is `gi_qtmd_staple_segments(W_index)`, which maps $[b_T, b_z, \eta, \perp]$ to the signed path segments

$$\left[\left(z, \eta + \frac{b_z}{2}\right), (\perp, b_T), \left(z, \frac{b_z}{2} - \eta\right)\right]. \tag{35}$$

The helper is covered by `tests/test_disconnected_gi_qtmd_staple_design.py`; it fixes the local, straight-PDF, fixed-length generic path, and invalid-index conventions.

The reference code-side helper is `create_fermion_TMD_GI(gauge, fermion, W_index)`. It applies the full staple path from the original local fermion field for each W-index, using repeated `gauge.pure_gauge.covDev` calls.

For production, the default `QTMD_1PT_GI_STAPLE_MODE=link_cache` first builds gauge-only staple transporters with `build_gi_qtmd_staple_links`. The construction converts an identity color matrix field into color-column fermions, applies the same `covDev` path, and converts the result back to a link field. This keeps the transporter convention identical to the reference direct-covariant derivative path while allowing all stochastic sources on the same gauge configuration to reuse the same staple links. The debug mode `QTMD_1PT_GI_STAPLE_MODE=direct_covdev` keeps the older source-by-source reference implementation.

The link-cache implementation is tested by:

1. identity-gauge equality with coordinate shifts;

2. nontrivial S8T32 test-gauge equality between cached links and direct `covDev`;

3. the PDF limit `GI_qTMD`$(b_T = 0, \eta = |b_z|/2) =$ `GI_PDF`(b_z);
4. a nonzero-transverse-staple HDF5 comparison at $(b_T, b_z, \eta) = (1, 2, 1)$, where `link_cache` and `direct_covdev` agree at the floating-point roundoff level.

The next physics validation should compare the local/PDF limits against the connected pion and proton qTMD workflows after constructing the disconnected ratio with matched two-point functions.